“Injury Risk Prediction System for Athletes Using Performance Data”

CO1 Assessment Tool 2

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DECLARATION

I, **DILLI BABU N 192521491,** of the **Department of B TECH IT**, Saveetha Institute of Medical and Technical Sciences, Saveetha University, Chennai, hereby declare that the Capstone Project Work entitled **“Injury Risk Prediction System for Athletes Using Performance Data”** is the result of my own bonafide efforts. To the best of my knowledge, the work presented herein is original, accurate, and has been carried out in accordance with principles of engineering ethics.

Place: Chennai
Date:

Signature of the Student

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1. Project Vision & Stakeholder Identification

1.1 Project Vision

The Injury Risk Prediction System is a cloud-based sports analytics platform that helps coaching and medical staff proactively reduce non-contact injuries among athletes. The system continuously ingests objective performance data (GPS/training load, heart rate, sprint counts), subjective wellness data (sleep, soreness, fatigue, stress), and historical injury records, then applies a machine-learning model to generate a daily injury-risk score for every athlete. The vision is to give sports science and medical teams an early-warning system — surfacing which athletes are at elevated risk of injury before it happens, so that training loads, recovery protocols, and match-day decisions can be adjusted in time to keep athletes healthy and available.

1.2 Project Objectives

- Continuously collect and validate training-load, biometric, and wellness data from wearables and manual entry.
- Calculate workload metrics such as the Acute:Chronic Workload Ratio (ACWR) to detect dangerous spikes in training load.
- Train and serve a machine-learning model that outputs a daily injury-risk score per athlete, with explainable contributing factors.
- Provide coaches and medical staff with a real-time, color-coded risk dashboard and automatic high-risk alerts.
- Maintain a complete injury history and return-to-play tracking system for every athlete.
- Ensure athlete health data is protected through granular access control, encryption, and regulatory-compliant privacy controls.

1.3 Key Stakeholders

Stakeholder	Role / Interest in the Project
Athletes (Primary Subjects)	Provide performance and wellness data; primary beneficiaries of reduced injury risk.
Coaches	Adjust training load and tactics based on daily risk scores and dashboards.
Strength & Conditioning Coaches	Design and modify conditioning programs in response to workload and risk alerts.
Sports Physiotherapists / Team Doctors	Review risk alerts, log injuries, manage rehabilitation and return-to-play decisions.
Sports Scientists / Performance Analysts	Monitor workload metrics (ACWR, GPS, RPE) and validate incoming data.
Data Scientists / ML Engineers	Build, train, and maintain the injury-risk prediction model.
Team Managers / Club Management	Review weekly risk and workload summary reports for squad planning.

Stakeholder	Role / Interest in the Project
Platform Administrator	Manages user accounts, data storage, integrations, and platform security.
Product Owner	Owns the product backlog, defines priorities, and represents stakeholder interests.
Scrum Master	Facilitates Agile ceremonies and removes impediments for the development team.
Development Team (Frontend/Backend/DevOps)	Designs, builds, tests, and deploys the platform's features.
UI/UX Designers	Design the dashboard, alert, and data-entry interfaces for usability.
QA / Test Engineers	Validate functionality, data accuracy, and reliability of risk predictions and alerts.

2. Epics and User Stories

The product requirements have been decomposed into seven Epics, each representing a major functional area of the platform. Every Epic is further broken down into well-structured User Stories following the standard Agile format: “As a <user>, I want <feature>, so that <benefit>.”

EPIC-01: User Accounts, Athlete Profiles & Team Management

Epic Goal: Allow coaches and staff to register, create athlete profiles, and invite team members with defined roles.

US-01: As a coach, I want to create an account and set up my team profile, so that I can start monitoring my athletes. *(Estimated: 3 pts)*

US-02: As a coach, I want to add athlete profiles with baseline biometric details (age, height, weight, position, injury history), so that the system has the data it needs for risk analysis. *(Estimated: 5 pts)*

US-03: As an admin, I want to invite team staff (physiotherapist, S&C coach, analyst) by email and assign roles, so that the right people have access to the right athlete data. *(Estimated: 5 pts)*

US-04: As an admin, I want to remove or change the role of a team member at any time, so that access stays aligned with current staff responsibilities. *(Estimated: 3 pts)*

EPIC-02: Performance & Training Data Ingestion

Epic Goal: Continuously capture objective and subjective athlete data as the foundation for risk prediction.

US-05: As a sports scientist, I want to import training-load data (GPS distance, heart rate, sprint counts) from wearable devices, so that the system has continuous, objective performance data. *(Estimated: 8 pts)*

US-06: As a coach, I want to manually log session RPE (Rate of Perceived Exertion) and session details, so that data gaps from missing or unavailable devices are filled. *(Estimated: 5 pts)*

US-07: As an athlete, I want to submit a daily wellness questionnaire (sleep quality, soreness, fatigue, stress), so that my subjective readiness is captured alongside objective data. *(Estimated: 5 pts)*

US-08: As a system, I want to auto-validate incoming data and flag incomplete or anomalous entries, so that the prediction model is not fed erroneous data. *(Estimated: 5 pts)*

EPIC-03: Injury Risk Prediction Engine

Epic Goal: Turn raw performance, wellness, and injury data into an actionable, explainable injury-risk score.

US-09: As a sports scientist, I want the system to automatically calculate each athlete's Acute:Chronic Workload Ratio (ACWR), so that dangerous spikes in training load are detected early. *(Estimated: 8 pts)*

US-10: As a data scientist, I want the system to run a trained ML model on combined performance, wellness, and historical injury data to generate a daily injury-risk score per athlete, so that high-risk athletes are proactively identified. *(Estimated: 13 pts)*

US-11: As a physiotherapist, I want to view the key factors contributing to an athlete's risk score, so that I can understand and act on the specific drivers of risk. *(Estimated: 8 pts)*

US-12: As a data scientist, I want to periodically retrain the prediction model with new injury outcome data, so that prediction accuracy keeps improving over time. *(Estimated: 8 pts)*

EPIC-04: Risk Alerts & Dashboard

Epic Goal: Surface risk information to coaches and medical staff in real time, in an actionable format.

US-13: As a coach, I want to see a color-coded dashboard of all athletes' current injury-risk levels, so that I can quickly identify who needs attention. *(Estimated: 8 pts)*

US-14: As a physiotherapist, I want to receive an automatic alert when an athlete's risk score crosses a high-risk threshold, so that I can intervene before an injury occurs. *(Estimated: 5 pts)*

US-15: As a coach, I want to view an individual athlete's risk-score trend over time on a graph, so that I can correlate risk spikes with training changes. *(Estimated: 5 pts)*

EPIC-05: Injury History & Medical Records

Epic Goal: Maintain a complete, structured record of injuries and recovery to inform both staff decisions and future predictions.

US-16: As a physiotherapist, I want to log details of an actual injury (type, date, severity, body part) when it occurs, so that the model can learn from real outcomes. *(Estimated: 5 pts)*

US-17: As a team doctor, I want to record and update an athlete's return-to-play status and rehabilitation milestones, so that coaching staff know when the athlete is available again. *(Estimated: 5 pts)*

US-18: As a physiotherapist, I want to view an athlete's complete injury-history timeline, so that I can factor past injuries into current risk assessment. *(Estimated: 3 pts)*

EPIC-06: Reporting & Export

Epic Goal: Allow risk and workload information to be shared and analyzed outside the platform.

US-19: As a team manager, I want to export a weekly team risk-and-workload summary report as a PDF, so that I can share it with club management. *(Estimated: 5 pts)*

US-20: As a sports scientist, I want to export raw performance and risk data as CSV, so that I can perform custom analysis outside the platform. *(Estimated: 3 pts)*

EPIC-07: Access Control, Privacy & Security

Epic Goal: Protect sensitive athlete health data with granular permissions and secure data handling.

US-21: As an athlete, I want to control who can view my personal health and biometric data, so that my medical privacy is protected. *(Estimated: 5 pts)*

US-22: As a platform administrator, I want to enforce encrypted storage and access logging for all athlete health data, so that the system remains secure and compliant with data-protection regulations. *(Estimated: 8 pts)*

3. Acceptance Criteria (Given–When–Then)

Clear, testable acceptance criteria have been written for the highest-priority user stories selected for the first sprint (and key stories from later sprints), using the Given–When–Then format to ensure each requirement is unambiguous and verifiable by QA.

US-02: Add an athlete profile

Given a logged-in coach on the team dashboard, when the coach clicks 'Add Athlete' and enters the athlete's biometric and position details,

Then, the system creates an athlete profile and makes it available for data logging and risk tracking.

Given the coach leaves a mandatory field (e.g., date of birth) blank, when the coach attempts to save the profile,

Then, the system blocks submission and highlights the missing required fields.

US-05: Import training-load data from wearables

Given a team's wearable devices are connected via the integration settings, when a training session ends and the device syncs,

Then, the athlete's GPS distance, heart-rate, and sprint-count data for that session are automatically imported and linked to the correct athlete and date.

Given the wearable sync fails or times out, when the import job runs,

Then, the system logs the failure and notifies the sports scientist so the session can be entered manually.

US-09: Calculate Acute:Chronic Workload Ratio (ACWR)

Given an athlete has at least 28 days of training-load data, when the nightly workload job runs,

Then, the system calculates the athlete's 7-day acute load and 28-day chronic load and derives the ACWR, updating it on the athlete's profile.

Given an athlete's ACWR exceeds the configured high-risk threshold (e.g., 1.5), when the calculation completes,

Then, the athlete is flagged for elevated workload risk on the dashboard.

US-10: Generate a daily injury-risk score

Given an athlete has current performance, wellness, and workload data available, when the daily prediction job runs,

Then, the ML model outputs a risk score (Low / Moderate / High) for that athlete, timestamped and stored for trend tracking.

Given required input data for an athlete is missing for the day, when the prediction job runs,

Then, the system marks that athlete's score as 'Insufficient Data' rather than generating an unreliable prediction.

US-13: View the team risk dashboard

Given a coach opens the team dashboard, when the page loads,

Then, every athlete is shown with a color-coded risk indicator (green/amber/red) and their name, position, and latest ACWR.

Given the coach filters by position or risk level, when a filter is applied,

Then, only athletes matching the filter are displayed, without a full page reload.

US-14: High-risk alert notification

Given an athlete's risk score changes from Moderate to High, when the daily prediction job completes,

Then, the assigned physiotherapist and coach receive an immediate in-app and email notification naming the athlete and top contributing risk factors.

Given an athlete remains at High risk for more than 3 consecutive days, when the daily job runs on day 3,

Then, an escalation notification is also sent to the team doctor.

US-16: Log an injury

Given a physiotherapist opens an athlete's profile after an injury occurs, when they select 'Log Injury' and enter type, date, severity, and body part,

Then, the injury record is saved, linked to the athlete's history, and included in the next model-retraining dataset.

Given an injury record is saved, when the athlete's current status is checked,

Then, the athlete is automatically marked 'Unavailable' on the team dashboard until return-to-play is confirmed.

US-21: Control visibility of personal health data

Given an athlete opens their privacy settings, when they restrict visibility of a data category (e.g., wellness responses) to medical staff only,

Then, coaching staff without medical clearance can no longer view that category, though risk scores derived from it remain visible.

Given a non-authorized user attempts to access restricted health data, when the request is made,

Then, the system denies access and logs the attempt in the audit trail.

4. Product Backlog Prioritization

The full product backlog was prioritized using a combination of MoSCoW categorization and a 1–10 Business Value score, taking into account athlete safety impact, technical dependencies, and the logical build order of the platform (e.g., accounts and data ingestion must exist before workload calculation or risk prediction are usable).

ID	Epic	User Story (short)	Value (1-10)	MoSCoW	Dependency
US-01	EPIC-01	create an account and set up my team profile…	9	Must	None
US-02	EPIC-01	add athlete profiles with baseline biometric details…	9	Must	US-01
US-03	EPIC-01	invite team staff and assign roles…	8	Must	US-01
US-05	EPIC-02	import training-load data from wearable devices…	10	Must	US-02
US-06	EPIC-02	manually log session RPE and session details…	7	Must	US-02
US-07	EPIC-02	submit a daily wellness questionnaire…	8	Must	US-02
US-08	EPIC-02	auto-validate and flag incomplete/anomalous data…	6	Should	US-05
US-09	EPIC-03	calculate Acute:Chronic Workload Ratio (ACWR)…	9	Must	US-05
US-10	EPIC-03	run ML model to generate a daily injury-risk score…	10	Must	US-09
US-13	EPIC-04	see a color-coded team risk dashboard…	9	Must	US-10
US-14	EPIC-04	receive an alert when risk crosses high threshold…	8	Should	US-10
US-11	EPIC-03	view key contributing factors behind a risk score…	7	Should	US-10
US-16	EPIC-05	log details of an actual injury…	7	Should	US-02
US-15	EPIC-04	view an athlete's risk trend over time…	6	Should	US-10
US-17	EPIC-05	record return-to-play status and rehab milestones…	6	Should	US-16
US-21	EPIC-07	control who can view my personal health data…	6	Should	US-02

ID	Epic	User Story (short)	Value (1-10)	MoSCoW	Dependency
US-22	EPIC-07	enforce encrypted storage and access logging…	6	Should	US-21
US-12	EPIC-03	periodically retrain the prediction model…	5	Could	US-10
US-18	EPIC-05	view an athlete's complete injury-history timeline…	5	Should	US-16
US-04	EPIC-01	remove or change the role of a team member…	4	Should	US-03
US-19	EPIC-06	export a weekly team risk/workload summary as PDF…	4	Could	US-13
US-20	EPIC-06	export raw performance and risk data as CSV…	3	Could	US-13

5. Sprint Backlog & Task Breakdown

Sprint 1 (2-week sprint) focuses on establishing the platform's foundation: team/account setup, athlete profiles, staff invitations, the data-ingestion pipeline (wearables, manual RPE, wellness), data validation, and the first workload metric (ACWR). These stories were selected because they are all rated 'Must' items and unblock the risk-prediction engine and dashboard planned for subsequent sprints.

5.1 Selected Sprint 1 Stories

ID	User Story	Story Points
US-01	As a coach, I want to create an account and set up my team profile...	3
US-02	As a coach, I want to add athlete profiles with baseline biometric details...	5
US-03	As an admin, I want to invite team staff and assign roles...	5
US-05	As a sports scientist, I want to import training-load data from wearables...	8
US-06	As a coach, I want to manually log session RPE and session details...	5
US-07	As an athlete, I want to submit a daily wellness questionnaire...	5
US-08	As a system, I want to auto-validate incoming data...	5
US-09	As a sports scientist, I want the system to calculate ACWR automatically...	8

5.2 Task Breakdown per Story

US-01: As a coach, I want to create an account and set up my team profile

- Design registration & login UI (Frontend)
- Implement email/password + OAuth (Google) sign-up API (Backend)
- Build team profile page (club name, sport, squad size)
- Write unit & integration tests for auth flow

US-02: As a coach, I want to add athlete profiles with baseline biometric details

- Design 'Add Athlete' form with biometric and position fields
- Define data schema for athlete profile and injury-history baseline
- Implement backend endpoint to create and store athlete profiles
- Add client- and server-side validation for mandatory fields

US-03: As an admin, I want to invite team staff and assign roles

- Design staff invite modal with role dropdown (Physio, S&C Coach, Analyst, Doctor)
- Implement email invitation service (send + accept link)
- Implement role-based access control (RBAC) middleware
- Test invite, accept, and role-enforcement scenarios

US-05: As a sports scientist, I want to import training-load data from wearable devices

- Integrate wearable-vendor API/SDK for session data pull
- Build data-mapping layer (device fields → platform schema)
- Implement scheduled sync job with retry-on-failure logic
- Test import against sample device data and simulated sync failures

US-06: As a coach, I want to manually log session RPE and session details

- Design manual session-entry form (RPE scale, duration, session type)
- Implement backend endpoint to store manually logged sessions
- Merge manual entries with device-imported sessions without duplication
- Unit test RPE calculation (session-RPE = RPE × duration)

US-07: As an athlete, I want to submit a daily wellness questionnaire

- Design mobile-friendly wellness questionnaire UI
- Implement backend storage of daily wellness responses
- Add daily reminder notification to athletes who haven't submitted
- Test questionnaire submission and duplicate-entry prevention

US-08: As a system, I want to auto-validate incoming data and flag anomalies

- Define validation rules (range checks, missing fields, duplicate sessions)
- Implement anomaly-flagging service for out-of-range values
- Build a data-quality review queue for sports scientists
- Test validation against known good and corrupted data samples

US-09: As a sports scientist, I want the system to calculate ACWR automatically

- Implement 7-day acute load and 28-day chronic load rolling calculations
- Implement ACWR derivation and threshold-based flagging
- Schedule nightly batch job to recompute ACWR for all athletes
- Validate calculation accuracy against manually computed test cases

6. Story Point Estimation Using Story Points

Story points were assigned using Planning Poker with the Fibonacci sequence (1, 2, 3, 5, 8, 13, 21), where the team estimated relative effort, complexity, and uncertainty rather than absolute time. The whole team (Product Owner, Scrum Master, Developers, QA, Data Scientist, and UX Designer) independently proposed estimates; divergent votes were discussed and re-voted until consensus was reached, following standard Planning Poker practice.

6.1 Estimation Scale Reference

Points	Relative Complexity Meaning
1	Trivial change, no unknowns (e.g., copy/label update).
2	Small, well-understood task with minimal integration.
3	Small feature with minor UI/backend work and clear requirements.
5	Medium feature involving both frontend and backend, some design decisions.
8	Complex feature with data-pipeline integration, multiple edge cases, or new architecture.
13	Very complex feature with significant unknowns, cross-cutting impact, or algorithmic/ML difficulty.

6.2 Story Point Justification (Sprint 1 Stories)

US-01: 3 pts — standard auth flow using well-known libraries; low uncertainty, minimal integration risk.

US-02: 5 pts — requires a new athlete data schema and validated form; moderate design decisions.

US-03: 5 pts — needs email invitation service plus role-based access control; moderate backend complexity.

US-05: 8 pts — third-party wearable API integration with scheduled sync and failure handling; high complexity and many edge cases.

US-06: 5 pts — builds on the session schema from US-05; moderate merge/deduplication logic.

US-07: 5 pts — mobile-friendly form plus reminder notifications; moderate frontend and backend work.

US-08: 5 pts — requires a rules engine and review queue; moderate new-architecture effort.

US-09: 8 pts — rolling-window statistical calculation with nightly batch processing and threshold logic; significant technical complexity and testing effort.

Total Sprint 1 Capacity Committed: 44 story points

This total falls within the team's assumed historical velocity range of 40–45 points per two-week sprint (for a 6-person cross-functional team including a data scientist), leaving a small buffer for code review, bug fixes, and sprint ceremonies.

7. Sprint Goal and Expected Deliverables

7.1 Sprint Goal

"Deliver a working foundation of the Injury Risk Prediction System where a coach can register a team, add athlete profiles, invite medical/S&C staff, and begin capturing training-load, RPE, and wellness data from day one — with the first automated workload metric (ACWR) calculated and validated — enabling an end-to-end demo of the data pipeline that will feed the injury-risk prediction engine in the next sprint."

7.2 Expected Sprint Deliverables

- Fully functional team registration, login, and staff-invitation module with role-based access.
- Athlete profile creation flow capturing baseline biometrics and injury-history fields.
- Wearable-device data-import pipeline with scheduled sync and failure handling.
- Manual session-logging (RPE) and daily athlete wellness-questionnaire modules.
- Data-validation and anomaly-flagging service with a review queue for sports scientists.
- Automated nightly ACWR calculation for every athlete with high-risk threshold flagging.
- A working end-to-end demo: an athlete's session and wellness data flowing in and producing an updated ACWR.

7.3 Definition of Done

- Code is peer-reviewed, merged to the main branch, and passes CI checks.
- Unit and integration tests written and passing for each story (minimum 80% coverage on new code).
- Acceptance criteria for each story verified by QA using the Given–When–Then scenarios in Section 3.
- Feature deployed to the staging environment and demonstrated at the Sprint Review.
- No critical or high-severity bugs open against the completed stories.

7.4 Sprint Timeline Overview

Day	Activity
Day 1	Sprint Planning — finalize sprint backlog and task assignments (this document).
Day 2–4	Accounts, athlete profiles, and staff-invitation development.
Day 5–7	Wearable data ingestion, manual session logging, and wellness questionnaire.
Day 8–9	Data validation/anomaly flagging and ACWR calculation service.
Day 10	Daily stand-ups throughout; mid-sprint backlog refinement.
Day 11–12	Integration testing, bug fixing, and QA verification against acceptance criteria.
Day 13	Sprint Review — demo to stakeholders.
Day 14	Sprint Retrospective — process improvement discussion.